

11(a)(i)	quantum of energy	M1
	of electromagnetic radiation	A1
11(a)(ii)	arrow (on Fig. 11.1) pointing upwards and to the right	B1
11(b)(i)	$\lambda = h / p$	C1
	$p = (6.63 \times 10^{-34}) / (544 \times 10^{-9})$	A1
	$= 1.22 \times 10^{-27} \text{ N s}$	
11(b)(ii)	energy = hc / λ	C1
	$= 6.63 \times 10^{-34} \times 3.00 \times 10^8 \times (540^{-1} - 544^{-1}) \times 10^9$	A1
	$= 2.7 \times 10^{-21} \text{ J}$	
11(c)	(smaller wavelength corresponds to) greater photon energy	B1
	any one point from: • (deflected) photon loses energy (so not possible) • (deflected) photon would need to gain energy (so not possible) • electron would need to lose energy (so not possible) • initially electron energy is zero (so not possible)	B1